EX. NO: 12 DATE: 16/09/2026

NMAP TO DISCOVER LIVE HOST USING ARP, ICMP SCAN, AND TCP/UDP PING SCAN IN PLATFORM

AIM:

Learn and implement Nmap host discovery techniques (ARP, ICMP, TCP, UDP ping) to identify live hosts on a network and compare their effectiveness.

PROCEDURE:

→ **ARP (LAN):** nmap -sn -PR 10.10.x.0/24
→ **ICMP Echo / Timestamp / Netmask:** nmap -sn -PE / -PP / -PM <target>
→ **TCP SYN / ACK (specify ports):** nmap -sn -PS22,80,443 / -PA80,443 <target>
→ **UDP Ping:** nmap -sn -PU53 10.10.x.0/24
→ **Combined / Disabled ARP:** nmap -sn -PE --disable-arp-ping <target>

INPUT (COMMANDS & PARAMETERS):

```
nmap -sn -PR 192.168.1.0/24 --packet-trace
nmap -sn -PE 10.10.x.0/24
nmap -sn -PP 10.10.x.100
nmap -sn -PM 10.10.x.100
nmap -sn -PS22,80,443 10.10.x.0/24
nmap -sn -PA80,443 10.10.x.0/24
nmap -sn -PU -PA -40 10.10.x.0/24
nmap -sn 10.10.x.0/24 --disable-arp-ping
```

- **Target IP range:** 10.10.x.100 or 10.10.x.0/24
- **Optional port lists:** for -PS, -PA, -PU (e.g., 22, 80, 443, 53)
- **Discovery Flags:** -sn, -PR, -PE, -PP, -PM, -PS, -PA, -PU, --disable-arp-ping

OUTPUT:

```
[ ARP Scan Output ]
Nmap scan report for 10.10.x.1 host
Host is up (0.00020s latency).
MAC Address: 02:08:75:5A:F2:24 (Gateway)
Nmap done: 256 IP addresses (2 hosts up)
[ ICMP Scan Output ]
Nmap scan report for 10.10.x.101 host
Host is up (0.00089s latency).
Nmap done: 1 IP address (1 host up)
```

RESULT:

The program was executed successfully.